

Class - 05

5 NUMBER SUMMARY

Analyzing Data

5 Number Summary

The **Five Number Summary** is a quick way to describe a dataset using **five special values**.

5 Number Summary provides a quick overview of the data's spread and center.

There are 5 key values -

- Minimum
- First Quartile(Q1)
- Median (Q2)
- Third Quartile(Q3)
- Maximum

REMOVING OUTLIERS

But First we have to see how to remove the outliers.

For Removing the outliers you need to have the upper fence and lower fence of your Data.

REMOVING OUTLIERS

The upper fence and lower fence formula is.

$$\text{upper fence} = Q3 + 1.5(IQR)$$

$$\text{lower fence} = Q1 - 1.5(IQR)$$

Lets see this with an example.

REMOVING OUTLIERS

[5, 7, 8, 10, 12, 13, 15, 18, 20, 21, 22, 24, 25, 28, 30, 32, 35, 40, 100]

- **Q1 (25th percentile): 12.5**
- **Q2 (Median): 21.0**
- **Q3 (75th percentile): 29.0**
- **$IQR = Q3 - Q1 = 29.0 - 12.5 = 16.5$**

REMOVING OUTLIERS

[5, 7, 8, 10, 12, 13, 15, 18, 20, 21, 22, 24, 25, 28, 30, 32, 35, 40, 100]

- **Q1 (25th percentile): 12.5**
 - **Q2 (): 21.0**
 - **Q3 (75th percentile): 29.0**
 - **$IQR = Q3 - Q1 = 29.0 - 12.5 = 16.5$**
-
- **Lower fence = $Q1 - 1.5 \times IQR = 12.5 - 24.75 = -12.25$**
 - **Upper fence = $Q3 + 1.5 \times IQR = 29.0 + 24.75 = 53.75$**

REMOVING OUTLIERS

[5, 7, 8, 10, 12, 13, 15, 18, 20, 21, 22, 24, 25, 28, 30, 32, 35, 40, ~~100~~]

- **Lower fence = $Q1 - 1.5 \times IQR = 12.5 - 24.75 = -12.25$**
- **Upper fence = $Q3 + 1.5 \times IQR = 29.0 + 24.75 = 53.75$**

Now you can clearly see which number is an outlier.

5 Number Summary

- **Minimum** → The smallest value in the data.
- **Q1 (First Quartile)** → The value at the 25% position
- **Median (Q2)** → The middle value (50% position).
- **Q3 (Third Quartile)** → The value at the 75% position
- **Maximum** → The largest value in the data.

Together, these 5 values summarize the distribution of the dataset.

→ 5 Number Summary gives a quick snapshot of the data's spread.

→ It helps to **detect outliers** or the values far away from Q1 and Q3.

5 Number Summary

Now let's see an example of 5 Number Summary

Suppose we have a dataset of exam marks :

10, 12, 14, 15, 18, 20, 22, 25, 30

Minimum = The smallest value = **10**

Maximum = The largest value = **30**

Median (Q2) = Middle value = **18**

Q1 = Median of the lower half (10, 12, 14, 15) = $\frac{(12 + 14)}{2} = \mathbf{13}$

Q3 = Median of the upper half (20, 22, 25, 30) = $\frac{(22 + 25)}{2} = \mathbf{23.5}$

5 Number Summary

So the **Five Number Summary** of the dataset is:

- Minimum = 10
- Maximum = 30
- Median(Q2) = 18
- Q1 = 13
- Q3 = 23.3

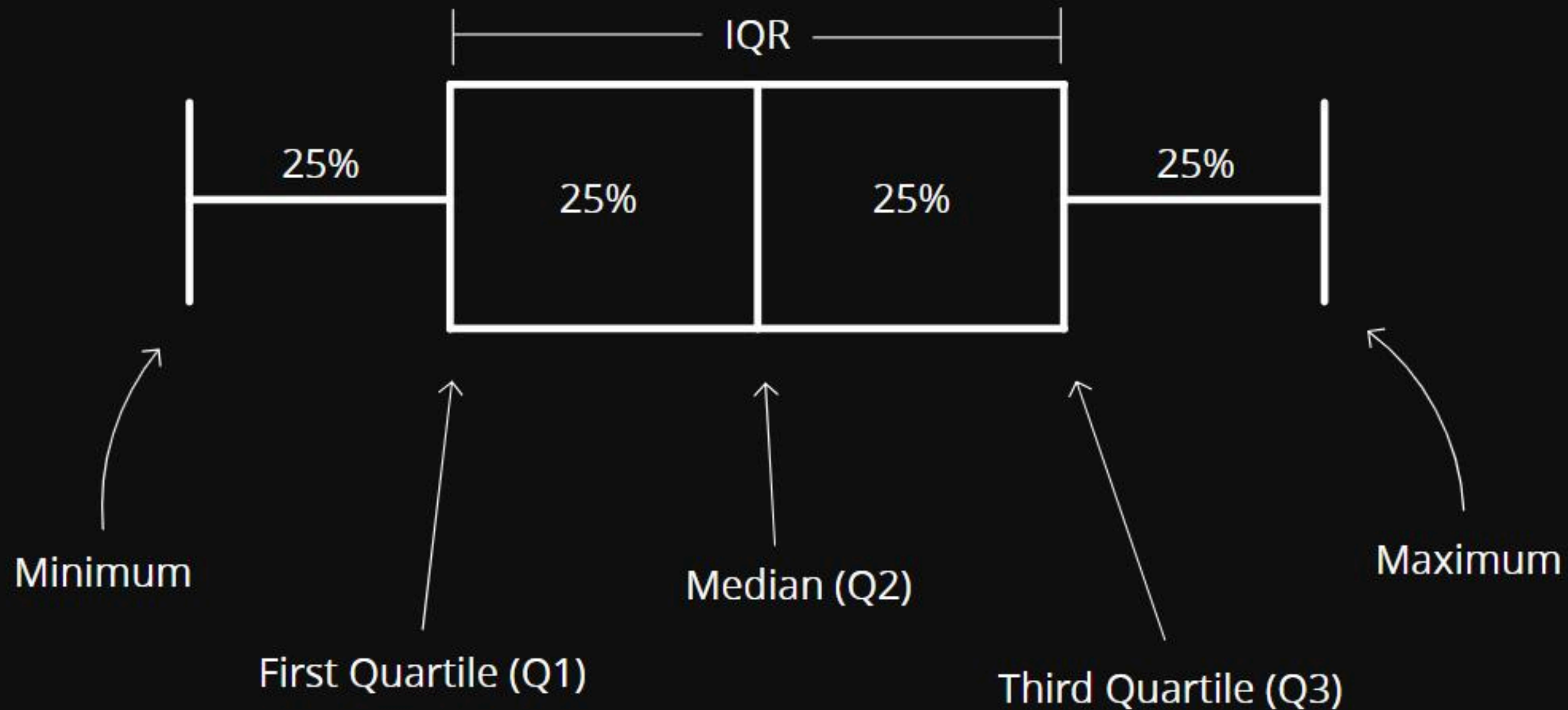
BOX PLOT

Analyzing Data

Box Plot

A **Box Plot** is a diagram representation that shows the **Five Number Summary** of data in one picture.

It helps us **see the Spread, the Center, and Outliers** very quickly.



Box Plot

Earlier we saw an example in 5 Number Summary

- Minimum = 10
- Maximum = 30
- Median(Q2) = 18
- Q1 = 13
- Q3 = 23.3

Now Let's create a Box Plot for this

